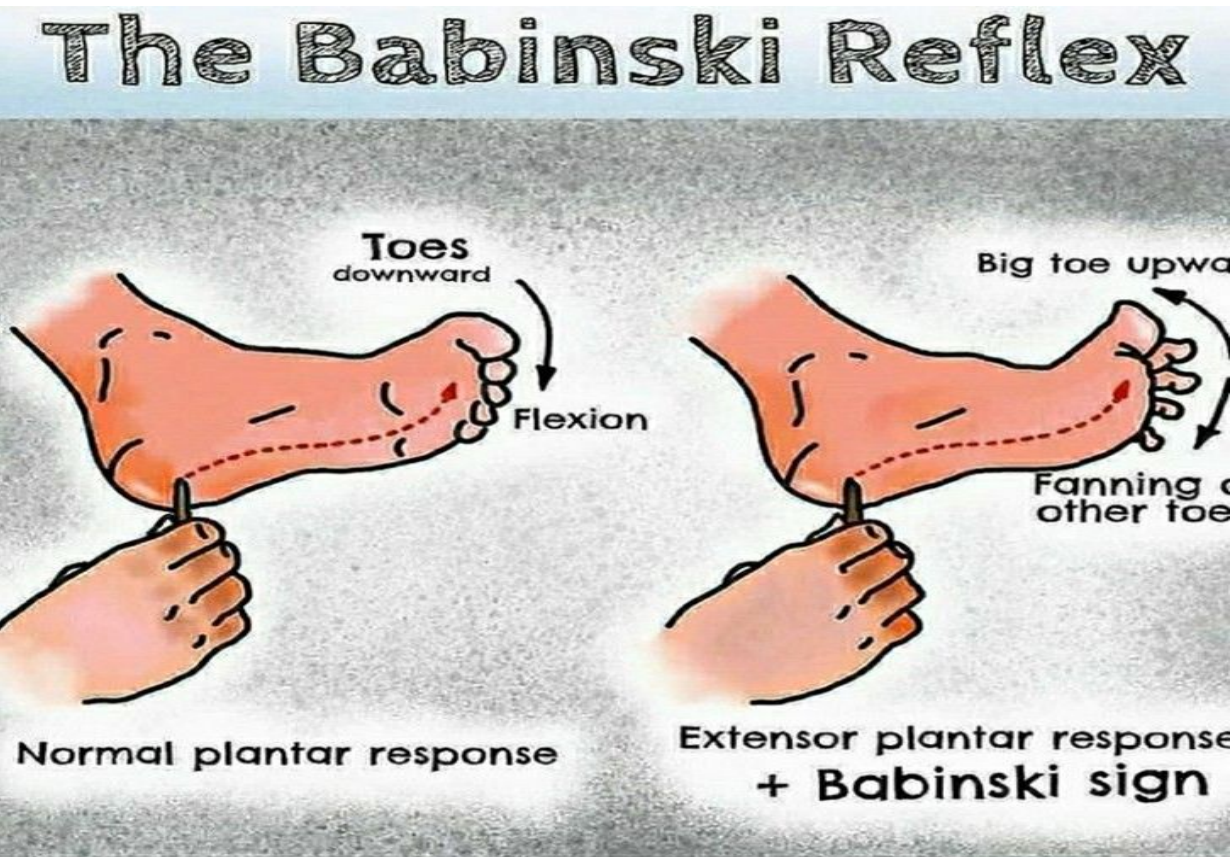


To Examine the Superficial and Deep reflexes



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THEORY

Reflex action is defined as an unconscious or automatic response to a stimulus.

A reflex can be classified on three bases:

- A. Time of appearance of reflex
- B. Location of receptors
- C. Number of synapses involved

A. Depending upon time of appearance, a reflex can be conditioned or unconditioned

UNCONDITIONED REFLEX	CONDITIONED REFLEX
Present since birth (congenital)	Acquired later in life
Fixed in number	Unlimited (habits)
Cerebral cortex is not required	Cerebral cortex is required
Experience not required	Experience is required
Always persists	Can be lost with time
e.g., corneal reflex, ankle reflex	e.g., driving, typing

B. Depending upon location of receptors, reflexes are of three types:

1. Superficial reflexes
2. Deep or tendon reflexes
3. Visceral reflexes.

C. Depending upon number of synapses:

1. Monosynaptic reflex: involving one synapse only e.g., stretch reflex
2. Polysynaptic reflex: involving multiple synapses e.g., withdrawal reflex

Superficial Reflexes

These are polysynaptic reflexes involving the CNS.

They are elicited by stimulation of the skin or mucous membrane and result in contraction of one or more muscles.

They are polysynaptic reflexes and are lost in lesions of the corticospinal tract or when the local sensory or motor arc is disrupted.

They are not dependent on the stimulation of tendons or muscle

Superficial reflexes

Superficial reflexes include the reflexes elicited by receptors present on the superficial surface of the body.

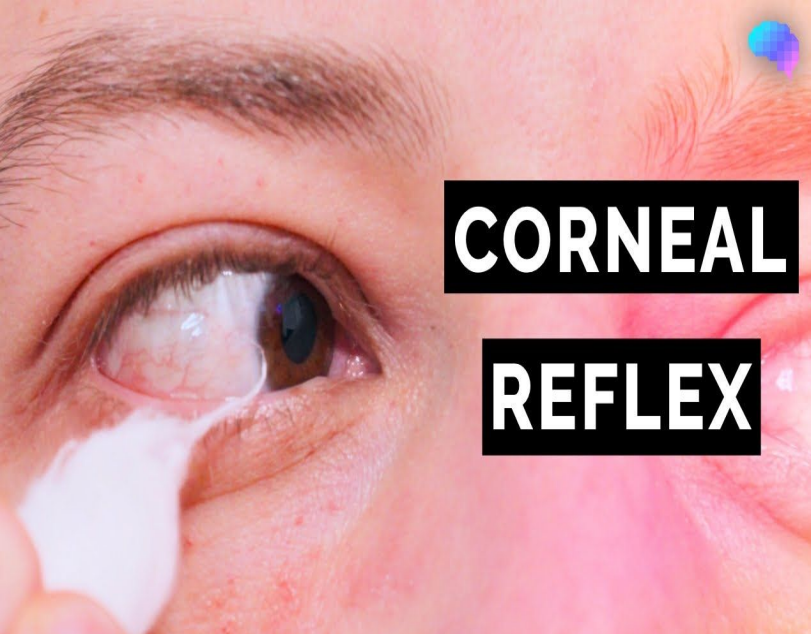
Sensory stimuli like touching or stretching the skin elicits a reflex action.

Examples of superficial reflexes are:

- 1) Conjunctival reflex
- 2) Corneal reflex
- 3) Pharyngeal reflex
- 4) Scapular reflex
- 5) Abdominal reflex
- 6) Plantar reflex.

EXPERIMENTAL PROCEDURE

S. No	Reflex	Procedure	Result / Response	Dermatome
1	Corneal/ Conjunctival reflex	Touch the cornea or conjunctiva with fine pointed end of cotton	Blinking of both eyes	C ₃ , C ₅
2	Pharyngeal reflex	Touch the back of tongue or roof of mouth with a wooden spatula	Reflex contraction of muscles of soft palate and pharynx	Cranial nerves 9, 10
3	Abdominal reflex	Scratch the abdominal skin below the level of umbilicus with a blunt object	Contraction of abdominal muscles	T ₈ -T ₁₂
4	Plantar reflex	Scratch the lateral surface of one foot from heel toward the big toe, then across the metatarsal medially with the blunt object.	Foot is dorsiflexed	L ₅ , S ₁
5	Babinski's reflex	Same manoeuvre as in plantar reflex	Great toe moves up & rest of the toes(fan)spread out. normal in infants<1year but abnormal in adults	L ₅ , S ₁
6	Scapular reflex	Expose the upper portion of the back. Stroke the skin in the interscapular region	Contraction of scapular muscles	C ₅₋₈ , T ₁



PRECAUTIONS:

- 1) To elicit the reflex, fine and gentle stimulus should be applied.
- 2) Person should be relaxed.
- 3) Both the sides of the body should be tested one by one.

Relevant Questions

Q-1-What is a reflex?

A reflex is an automatic, involuntary physical action that occurs almost instantly in response to a stimulus without conscious thought.

Q-2-What are the components of reflex arc?

- A reflex arc is a neural pathway consisting of five components:
- receptor,
- sensory neuron,
- integration center,
- motor neuron,
- and effector.
- It is responsible for the automatic response of the body to a stimulus.

Q-3-What is Babinski's extensor plantar response? What is its significance?

Babinski's extensor plantar response, or the Babinski sign, is an abnormal upward extension of the big toe in response to stroking the sole of the foot.

This response is considered normal in infants and toddlers but is considered a significant indicator of an upper motor neuron lesion (damage to the brain or spinal cord) in adults.

Its presence suggests damage to the corticospinal tract, the nerve pathway that controls voluntary movement

Q-4-What are the physiological conditions in which Babinski's sign becomes positive?

A positive Babinski sign in an adult indicates central nervous system (CNS) damage, particularly to the corticospinal tract, and can be a sign of diseases like stroke, multiple sclerosis (MS), ALS, brain tumors, meningitis, and spinal cord injuries. The sign involves the great toe pointing upward and the other toes splaying out when the sole of the foot is stroked, which is an abnormal response in adults, as the normal response is for the toes to curl downward

Q-5-What clinical information can be obtained by eliciting the superficial reflexes in human?

Eliciting superficial reflexes provides clinical information about the integrity of the central nervous system (CNS), including the spinal cord, brainstem, and their connections to the peripheral nervous system.

Specifically, it can help localize neurological lesions by revealing the level of a spinal cord injury or upper motor neuron lesion.

Abnormalities in reflexes like the plantar reflex can indicate problems with the corticospinal tracts while the abdominal and cremasteric reflexes can assess the function of the thoracic and lumbar spinal cord segments, respectively.

Q-6-What are the differences between conditional and unconditional reflexes?

Unconditioned reflexes are automatic, innate responses to a stimulus that do not require learning, such as blinking when something enters your eye. Conditioned reflexes are learned behaviors that develop over an individual's life by forming an association between a neutral stimulus and a natural unconditioned stimulus, like Pavlov's dogs salivating at the sound of a bell after it's paired with food.

Q-7-Identify the reflexe shown in figure 2.4

Pupillary Reflex

Q-8-Write the steps of performance of this reflex?

Gently point the focal light into one eye, this is known as the direct pupillary light reflex. Then, withdraw the light for few seconds, followed by stimulating the same eye again but this time observe the indirect, or consensual, PLR in the opposite eye. It may be helpful to have the nurse control the light stimulus while you observe the unstimulated eye.

Q-9-What is the expected response of this reflex?

The pupillary light reflex constricts the pupil in response to light, and pupillary constriction is achieved through the innervation of the iris sphincter muscle.

Q-10-Which type of reflex is this?

There are two primary types of pupillary reflexes: the pupillary light reflex, where the pupil constricts in bright light and dilates in darkness, and the pupillary accommodation reflex, where the pupil constricts when focusing on a near object. The "direct" and "consensual" reflexes describe how the light reflex is observed: direct is in the eye receiving the light, while consensual is in the other eye

Q-11-Is this reflex present by birth or acquired later?

The pupil light reflex has been reported to be absent in premature infants of <30 weeks gestational age, but developing gradually and is well developed by 35 weeks based on previous reported studies. 2,4 Another study reported that a light response of 0.5 mm or more can be observed from at >31 weeks of age.

Q-12-Name two other superficial reflexes?

Two other superficial reflexes are the corneal reflex (involuntary blinking when the cornea is stimulated) and the plantar reflex (toe movement in response to scratching the sole of the foot). Other examples include the abdominal reflex and the cremasteric reflex.

Practical 2

Objective: To elicit the Deep Reflexes in a human subject

PREREQUISITES: Hammer and torch

THEORY

The contraction of the muscle in response to a sudden stretch produced by striking the tendon is called a tendon reflex.

The tendon reflexes are stretch reflexes because they are elicited by stretching the muscle.

Since the receptors are located deep in muscles, they are also called deep reflexes. Deep reflexes include

- Biceps jerk
- Triceps jerk
- Supinator jerk
- knee jerk
- Ankle jerk

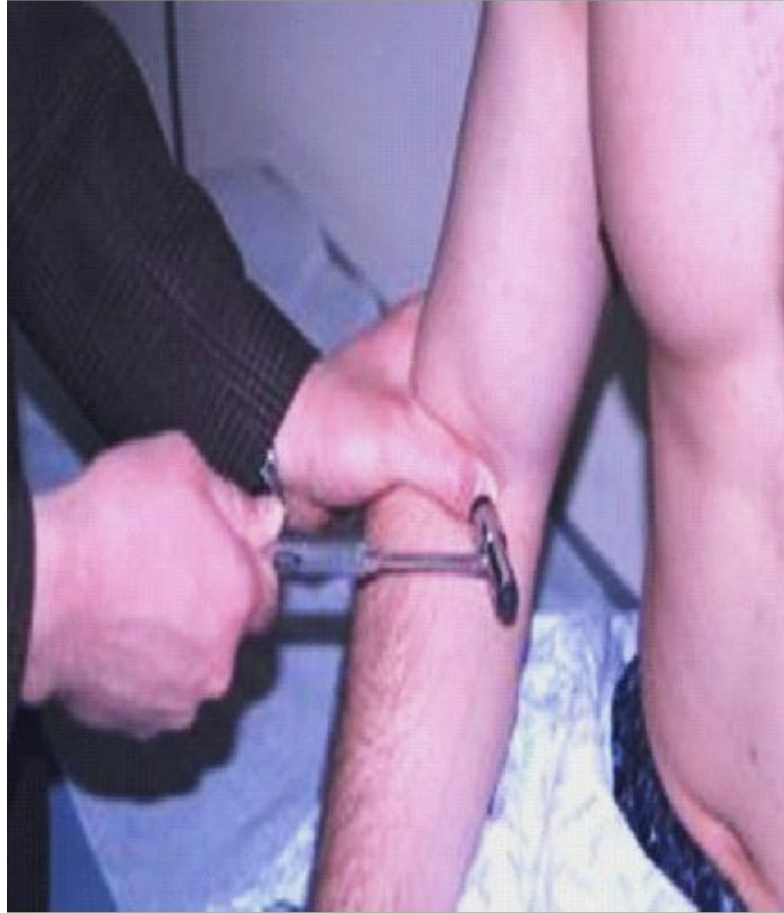
CLINICAL IMPORTANCE

- Reflexes may be normal, increased or diminished.
- Increased (Brisk) tendon jerks are the sign of upper motor neuron lesion if present in one limb or one half of the body. They are also seen in anxiety, tetanus, and thyrotoxicosis.
- Reflexes are diminished or absent if there is lower motor neuron lesion, as in neuropathy, spinal shock, and coma. Reflexes are usually normal in myopathies.
- Reflexes are increased in amplitude when muscle being tested is contracted forcefully known as Reinforcement of reflexes. This is performed by strong voluntary muscular effort of subject.
- The involuntary muscular contraction and relaxation evoked by a sudden stretch of muscles is known as clonus. Sustained clonus is the one that continues as long as the stretch is applied; this is a sign of upper motor neuron lesion and is associated with brisk reflexes.

EXPERIMENTAL PROCEDURE:

S.NO	Name of Reflex	Procedure to elicit the reflex	Results	Dermatome involved
1	Knee jerk (Sitting position)	Strike the patellar tendon directly with the help of patellar hammer.	Contraction of quadriceps and extension at knee.	L2, L3, L4
2	Ankle jerk (Supine position)	Place the lower limb in such a way that it is everted and slightly flexed. Strike the Achilles tendon with patellar hammer.	Flexion at the ankle joint due to contraction of calf muscles.	S1, S2
3	Biceps jerk (Sitting position)	Flex the elbow and place forearm in semi-pronated position, feel the biceps with the thumb and strike the patellar hammer on the thumb.	Contraction of biceps and flexion at elbow joint.	C5, C6
4	Triceps jerk (Sitting position)	Flex the subject's arm at elbow and allow forearm to rest on his/her abdomen. Tap the triceps tendon directly above the olecranon.	Contraction of triceps and extension at elbow joint.	C6, C7, C8
5	Supinator or Brachioradialis jerk	Hold the hand of the subject and slightly stretch the brachioradialis muscle by laterally bending the hand in the opposite direction. Strike the radius 1-2 inches above the wrist over its styloid process.	Flexion at the elbow and supination of the forearm.	C5, C6

BICEPS JERK



KNEE JERK



ANKLE JERK



PRECAUTIONS:

- 1-Explain the procedure to the subject before eliciting the reflex
- 2-Subject should be relaxed and part of body to be tested should be exposed
- 3-Concentrate on the movements of the muscle to be tested not the movement of the joint
- 4-Make sure that there is no joint disease.

OBSERVATION:

S. No	Name of reflex	Right	left
1.	Biceps reflex		
2.	Triceps reflex		
3.	Supinator reflex		
4.	Knee jerk		
5.	Ankle jerk		

Relevant Questions

Q-1-What is the significance of the Phenomenon of reinforcement?

Reinforcement is a fundamental aspect of operant conditioning that is used to describe the strengthening of a situation or element. In behavioral psychology, the term refers to the application of any stimulus which facilitates and increases the likelihood of a specific response occurring

Q-2-What are the causes of hyper reflexia?

Hyperreflexia is caused by conditions affecting the central nervous system, such as spinal cord injuries, stroke, Parkinson's disease, multiple sclerosis (MS), and trauma. It can also result from nerve compression, certain infections, metabolic issues like hyperthyroidism, some medications (like SSRIs), and in some cases, anxiety or withdrawal from substances such as alcohol.

Q-3-What are different abnormalities of tendon reflex?

Abnormalities of tendon reflexes include areflexia (absent reflex), hyporeflexia (diminished reflex), and hyperreflexia (exaggerated reflex), which can include clonus.

Q-4-Write the procedure to elicit biceps jerk?

Place your thumb firmly over the biceps tendon, with your fingers curling around the elbow, and tap briskly. The forearm will flex at the elbow.

Q-5-Which type of reflex is this?

Biceps reflex is a deep tendon reflex (DTR) test (also known as a muscle-stretch reflex test) that examines the function of the C5 reflex arc and the C6 reflex arc. The test is performed by using a tendon hammer to quickly depress the biceps brachii tendon as it passes through the cubital fossa.

Q-6-Which dermatomes are involved in this reflex?

C5 – Flexion at the elbow, biceps.

C6 – Flexion at the elbow, brachioradialis.

C7 – Extension at the elbow, triceps.

C8 – Finger flexion

Q-7-Name the receptors involved in this reflex?

Muscle involved: biceps brachii.

Nerve supply: musculocutaneous.

Segmental innervation: C5-C6

ame the receptors involved in this reflex